Theory of Commutative Algebra 6

Theory Notes

July 29, 2026

This file is a professionally reorganized and cleaned version.

How to structure for `report`:

- Use `\chapter{...}` for big blocks (major topics).
- Use `\section{...}` and `\subsection{...}` inside chapters.
- Avoid starred forms (`\section*` etc.) if you want entries in the Table of Contents.
- Compile **twice** so the Table of Contents is filled.

Quick conversion tip (TeXstudio Find/Replace):

- Replace `\section*{` with `\section{`
- Replace `\subsection*{` with `\subsection{`
- For large top-level blocks, replace the first `\section{` with `\chapter{`

If you insist on unnumbered headings that still appear in the ToC, use the helpers: `\chaptertoc`, `\sectiontoc`, `\subsectiontoc`.

Contents

Chapter 1

Examples and intuition for $D(A)$

1.1 Recall the definition and the slogan

1.2 Domain case: $A = k[t]$

1.3 Two crossing components: $A = k[x, y]/(xy)$

1.4 Nilpotent example: $A = k[t]/(t^2)$

1.5 Localization removes primes that meet S

1.6 Embedded associated prime: $A = k[x, y]/(x^2, xy)$

1.7 Product ring example: $A = k \times k$

1.8 Many zero divisors, one prime

1.9 Relation with $\mathrm{Ass}(A)$ and zero divisors

1.10 How to compute $D(A)$ in practice

1.11 Final summary

Chapter 2

Concrete polynomial and ring examples for $D(A)$

Chapter 3

Classes in quotient rings, minimal primes, and associated primes

Chapter 4

Concrete annihilators: $\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, polynomial rings, and quotient rings

Chapter 5

Annihilators, minimal primes, and associated primes for modules: concrete examples

Chapter 6

Concrete residue classes in $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$, $k[x]/(x)$, $A[t]/(t)$, and $k[x, y]/(xy)$